

Last login: Mon Sep 21 12:31:53 on ttys003
/var/folders/g6/2h7p_f2x2qbcf0_rwd7zs4y00000gn/T/orbstack-open-terminal_8410583A.sh ; exit;
harsha@Harshas-MacBook-Air-3 ~ % /var/folders/g6/2h7p_f2x2qbcf0_rwd7zs4y00000gn/T/orbstack-open-terminal_8410583A.sh ; exit;

Orbstack update available!
Run "orb update" to update.
Updates include improvements, features, and fixes.

```
harsha@Harsha:~$ mkdir shell_parser
harsha@Harsha:~$ cd shell_parser
harsha@Harsha:~/shell_parser$ pwd
/home/harsha/shell_parser
harsha@Harsha:~/shell_parser$ nano tokenizer.c
harsha@Harsha:~/shell_parser$ gcc -Wall -Wextra tokenizer.c -o tokenizer
harsha@Harsha:~/shell_parser$ ./tokenizer
Enter command: r
```

```
--- Token Stream ---
0: WORD : r
Token stream is valid.
harsha@Harsha:~/shell_parser$ ls -l | grep .c
-rw-r--r-- 1 harsha harsha 3734 Sep 21 12:37 tokenizer.c
harsha@Harsha:~/shell_parser$ ./tokenizer
Enter command: Enter command: ls -l | grep .c
```

```
--- Token Stream ---
0: WORD : ls
1: WORD : -l
2: PIPE : |
3: WORD : grep
4: WORD : .c
```

Token stream is valid.

```
--- Token Stream ---
0: WORD : Enter
1: WORD : command:
2: WORD : ls
3: WORD : -l
4: PIPE : |
5: WORD : grep
6: WORD : .c
```

Token stream is valid.

```
harsha@Harsha:~/shell_parser$
harsha@Harsha:~/shell_parser$ --- Token Stream ---
-bash: ---: command not found
harsha@Harsha:~/shell_parser$ 0: WORD : ls
-bash: 0: command not found
harsha@Harsha:~/shell_parser$ 1: WORD : -l
-bash: 1: command not found
harsha@Harsha:~/shell_parser$ 2: PIPE : |
> 3: WORD : grep
-bash: 3: command not found
-bash: 2: command not found
harsha@Harsha:~/shell_parser$ 4: WORD : .c
-bash: 4: command not found
harsha@Harsha:~/shell_parser$
harsha@Harsha:~/shell_parser$ Token stream is valid.
-bash: Token: command not found
harsha@Harsha:~/shell_parser$ ./tokenizer
Enter command: cat < input.txt > output.txt
```

```
--- Token Stream ---
0: WORD : cat
1: INPUT : <
2: WORD : input.txt
3: OUTPUT : >
4: WORD : output.txt
Token stream is valid.
harsha@Harsha:~/shell_parser$ ls -l | grep .c
-rw-r--r-- 1 harsha harsha 3734 Sep 21 12:37 tokenizer.c
```

```
harsha@Harsha:~/shell_parser$ 4: WORD      : .c
-bash: 4:: command not found
harsha@Harsha:~/shell_parser$
harsha@Harsha:~/shell_parser$ Token stream is valid.
-bash: Token: command not found
harsha@Harsha:~/shell_parser$ ./tokenizer
Enter command: cat < input.txt > output.txt
```

```
--- Token Stream ---
0: WORD      : cat
1: INPUT      : <
2: WORD      : input.txt
3: OUTPUT     : >
4: WORD      : output.txt
Token stream is valid.
harsha@Harsha:~/shell_parser$ ls -l | grep .c
-rw-r--r-- 1 harsha harsha 3734 Sep 21 12:37 tokenizer.c
harsha@Harsha:~/shell_parser$ ls
-bash: syntax error near unexpected token `|'
harsha@Harsha:~/shell_parser$ ls | grep
tokenizer tokenizer.c
harsha@Harsha:~/shell_parser$ nano parser.c
harsha@Harsha:~/shell_parser$ gcc -Wall -Wextra parser.c -o parser
harsha@Harsha:~/shell_parser$ gcc -Wall -Wextra parser.c -o parser
harsha@Harsha:~/shell_parser$ ./parser
Enter command: ls -l
```

Syntax is valid.

```
===== PARSE TREE =====
PIPELINE
|
+--- COMMAND 1
      +--- ARG: ls
      +--- ARG: -l
```

```
===== EXECUTION STRUCTURE =====
Command 1:
  argv: [ls] [-l]
harsha@Harsha:~/shell_parser$ ./parser
Enter command: ls -l | grep .c
```

Syntax is valid.

```
===== PARSE TREE =====
PIPELINE
|
+--- COMMAND 1
      +--- ARG: ls
      +--- ARG: -l
|
+--- COMMAND 2
      +--- ARG: grep
      +--- ARG: .c
```

```
===== EXECUTION STRUCTURE =====
Command 1:
  argv: [ls] [-l]
Command 2:
  argv: [grep] [.c]
harsha@Harsha:~/shell_parser$ ./parser
Enter command: Enter
```

Syntax is valid.

```
===== PARSE TREE =====
PIPELINE
|
+--- COMMAND 1
      +--- ARG: Enter
```

```
===== EXECUTION STRUCTURE =====
Command 1:
  argv: [Enter]
harsha@Harsha:~/shell_parser$
```

